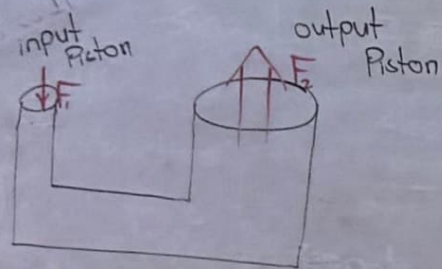


any change in pressure

Pascal Principle



Any change in a pressure will be transmitted equally to every point in a container

$$P_1 = P_2 \rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Using: it is used to تکبیر magnify a force.

Applications: ① Car breaks, ② Car lifters.

Conditions:

- ① Fluid is at rest
- ② Container is closed

ex. 5

In Pascal experiment, The two Pistons having areas of A_1 25 cm² and A_2 750 cm².
if the total weight on the output piston is 15000 N. The force on the input piston is:

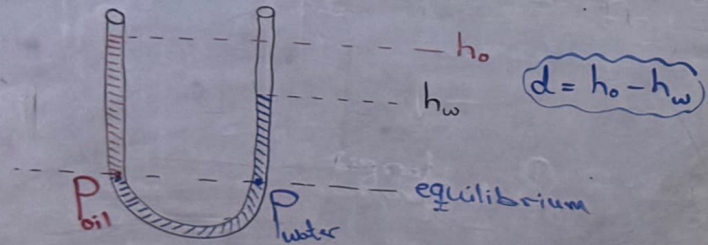
$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_1 = F_2 \frac{A_1}{A_2} = 15000 \times \frac{25}{750} = 500 \text{ N}$$

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ex. 4

U-shaped tube contains water and oil. The density of oil is:



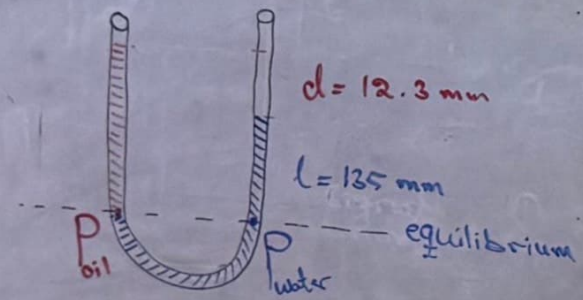
$\therefore$ static equilibrium: $P_o = P_w$

$$\rho_o g h_o = \rho_w g h_w$$

$$\rho_o h_o = \rho_w h_w$$

$$\rho_o < \rho_w$$

$$h_o > h_w$$



∴ static equilibrium: $P_o = P_w$

$$\rho_o g h_o = \rho_w g h_w$$

$$\rho_o h_o = \rho_w h_w$$

$$\rho_o < \rho_w$$

$$h_o > h_w$$

ex. 4

U-shaped tube contains water and oil. The density of oil is:

$$h_w = 135 \text{ mm}, h_o = 135 + 12.3 = 147.3 \text{ mm}$$

$$\therefore \rho_o h_o = \rho_w h_w$$

$$\rho_o = \rho_w \times \frac{h_w}{h_o} = 1000 \times \frac{135}{147.3} = 916.5 \text{ kg/m}^3$$

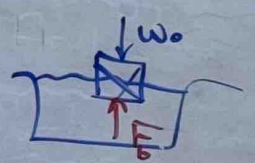
لیکچر ۱۱ - فزکس II

$\omega_o \rightarrow$ object weight

$\omega_f \rightarrow$ displaced fluid

$F_b \rightarrow$ Buoyant force

① Floating: rest $\rightarrow a = 0$
 Partially immersed
 $V_o \neq V_f$



$$\omega_o = F_b$$

$$m_o = m_f$$

$$\rho_o V_o = \rho_f V_f$$

$$\rho_o A_o h_o = \rho_f A_f h_f$$

$$\rho_o h_o = \rho_f h_f \quad \text{floating}$$

$$\rho_o < \rho_f$$

$$h_o > h_f$$